Captstone Data 205

Charanpreet Singh

2026-04-20

#Librarires Laoding

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.5.1     ✔ tibble    3.2.1
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.0.4
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
library(lubridate)
library(tidycensus)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

#ZHVI County Csv

```
housing <- read.csv(
  "~/Downloads/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv",
  check.names = FALSE
)
```

#Zillow City data

```
housing_city <- read.csv(
  "~/Downloads/City_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv",
  check.names = FALSE
)
```

#Cencus API key

```
census_api_key("YOUR KEY HERE",overwrite=TRUE)
```

```
## To install your API key for use in future sessions, run this function with `install =
TRUE`.
```

#Filter Montgomery County cities

```
#Creating variable to store all of MoCO cities into
moco_cities <- housing_city %>%
  filter(StateName == "MD", CountyName == "Montgomery County")

#outputting all cities in MoCo
unique(moco_cities$RegionName)
```

```
##  [1] "Silver Spring"      "Rockville"          "Gaithersburg"
##  [4] "Germantown"         "Bethesda"           "Potomac"
##  [7] "Montgomery Village" "Takoma Park"        "Chevy Chase"
## [10] "North Potomac"      "Olney"              "Derwood"
## [13] "Clarksburg"         "Burtonsville"       "Kensington"
## [16] "Damascus"           "Boyds"              "Brookeville"
## [19] "Darnestown"         "Poolesville"        "Sandy Spring"
## [22] "Dickerson"          "Cabin John"         "Ashton"
## [25] "Chevy Chase View"   "Garrett Park"       "Laytonsville"
## [28] "Washington Grove"   "Spencerville"       "Brinklow"
## [31] "Glen Echo"
```

#Data inspection

```
glimpse(moco_cities)
```

```
## Rows: 31
## Columns: 323
## $ RegionID     <int> 7081, 33714, 45383, 31732, 37406, 47210, 19445, 47942, 39…
## $ SizeRank     <int> 79, 421, 425, 587, 652, 1289, 1883, 2438, 2519, 2520, 271…
## $ RegionName   <chr> "Silver Spring", "Rockville", "Gaithersburg", "Germantown…
## $ RegionType   <chr> "city", "city", "city", "city", "city", "city", "city", "…
## $ StateName    <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD…
## $ State        <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD…
## $ Metro        <chr> "Washington-Arlington-Alexandria, DC-VA-MD-WV", "Washingt…
## $ CountyName   <chr> "Montgomery County", "Montgomery County", "Montgomery Cou…
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## $ `2026-02-28` <dbl> 550772.3, 618933.4, 528006.9, 423911.1, 1150415.5, 138759…
## $ `2026-03-31` <dbl> 550712.4, 618892.3, 527608.4, 423001.5, 1152218.6, 139021…
```

```
colnames(moco_cities)
```

```
##   [1] "RegionID"   "SizeRank"   "RegionName" "RegionType" "StateName"
##   [6] "State"      "Metro"      "CountyName" "2000-01-31" "2000-02-29"
##  [11] "2000-03-31" "2000-04-30" "2000-05-31" "2000-06-30" "2000-07-31"
##  [16] "2000-08-31" "2000-09-30" "2000-10-31" "2000-11-30" "2000-12-31"
##  [21] "2001-01-31" "2001-02-28" "2001-03-31" "2001-04-30" "2001-05-31"
##  [26] "2001-06-30" "2001-07-31" "2001-08-31" "2001-09-30" "2001-10-31"
##  [31] "2001-11-30" "2001-12-31" "2002-01-31" "2002-02-28" "2002-03-31"
##  [36] "2002-04-30" "2002-05-31" "2002-06-30" "2002-07-31" "2002-08-31"
##  [41] "2002-09-30" "2002-10-31" "2002-11-30" "2002-12-31" "2003-01-31"
##  [46] "2003-02-28" "2003-03-31" "2003-04-30" "2003-05-31" "2003-06-30"
##  [51] "2003-07-31" "2003-08-31" "2003-09-30" "2003-10-31" "2003-11-30"
##  [56] "2003-12-31" "2004-01-31" "2004-02-29" "2004-03-31" "2004-04-30"
##  [61] "2004-05-31" "2004-06-30" "2004-07-31" "2004-08-31" "2004-09-30"
##  [66] "2004-10-31" "2004-11-30" "2004-12-31" "2005-01-31" "2005-02-28"
##  [71] "2005-03-31" "2005-04-30" "2005-05-31" "2005-06-30" "2005-07-31"
##  [76] "2005-08-31" "2005-09-30" "2005-10-31" "2005-11-30" "2005-12-31"
##  [81] "2006-01-31" "2006-02-28" "2006-03-31" "2006-04-30" "2006-05-31"
##  [86] "2006-06-30" "2006-07-31" "2006-08-31" "2006-09-30" "2006-10-31"
##  [91] "2006-11-30" "2006-12-31" "2007-01-31" "2007-02-28" "2007-03-31"
##  [96] "2007-04-30" "2007-05-31" "2007-06-30" "2007-07-31" "2007-08-31"
## [101] "2007-09-30" "2007-10-31" "2007-11-30" "2007-12-31" "2008-01-31"
## [106] "2008-02-29" "2008-03-31" "2008-04-30" "2008-05-31" "2008-06-30"
## [111] "2008-07-31" "2008-08-31" "2008-09-30" "2008-10-31" "2008-11-30"
## [116] "2008-12-31" "2009-01-31" "2009-02-28" "2009-03-31" "2009-04-30"
## [121] "2009-05-31" "2009-06-30" "2009-07-31" "2009-08-31" "2009-09-30"
## [126] "2009-10-31" "2009-11-30" "2009-12-31" "2010-01-31" "2010-02-28"
## [131] "2010-03-31" "2010-04-30" "2010-05-31" "2010-06-30" "2010-07-31"
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## [156] "2012-04-30" "2012-05-31" "2012-06-30" "2012-07-31" "2012-08-31"
## [161] "2012-09-30" "2012-10-31" "2012-11-30" "2012-12-31" "2013-01-31"
## [166] "2013-02-28" "2013-03-31" "2013-04-30" "2013-05-31" "2013-06-30"
## [171] "2013-07-31" "2013-08-31" "2013-09-30" "2013-10-31" "2013-11-30"
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## [181] "2014-05-31" "2014-06-30" "2014-07-31" "2014-08-31" "2014-09-30"
## [186] "2014-10-31" "2014-11-30" "2014-12-31" "2015-01-31" "2015-02-28"
## [191] "2015-03-31" "2015-04-30" "2015-05-31" "2015-06-30" "2015-07-31"
## [196] "2015-08-31" "2015-09-30" "2015-10-31" "2015-11-30" "2015-12-31"
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## [211] "2016-11-30" "2016-12-31" "2017-01-31" "2017-02-28" "2017-03-31"
## [216] "2017-04-30" "2017-05-31" "2017-06-30" "2017-07-31" "2017-08-31"
## [221] "2017-09-30" "2017-10-31" "2017-11-30" "2017-12-31" "2018-01-31"
## [226] "2018-02-28" "2018-03-31" "2018-04-30" "2018-05-31" "2018-06-30"
## [231] "2018-07-31" "2018-08-31" "2018-09-30" "2018-10-31" "2018-11-30"
## [236] "2018-12-31" "2019-01-31" "2019-02-28" "2019-03-31" "2019-04-30"
## [241] "2019-05-31" "2019-06-30" "2019-07-31" "2019-08-31" "2019-09-30"
## [246] "2019-10-31" "2019-11-30" "2019-12-31" "2020-01-31" "2020-02-29"
## [251] "2020-03-31" "2020-04-30" "2020-05-31" "2020-06-30" "2020-07-31"
## [256] "2020-08-31" "2020-09-30" "2020-10-31" "2020-11-30" "2020-12-31"
```

```
## [261] "2021-01-31" "2021-02-28" "2021-03-31" "2021-04-30" "2021-05-31"
## [266] "2021-06-30" "2021-07-31" "2021-08-31" "2021-09-30" "2021-10-31"
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## [281] "2022-09-30" "2022-10-31" "2022-11-30" "2022-12-31" "2023-01-31"
## [286] "2023-02-28" "2023-03-31" "2023-04-30" "2023-05-31" "2023-06-30"
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## [296] "2023-12-31" "2024-01-31" "2024-02-29" "2024-03-31" "2024-04-30"
## [301] "2024-05-31" "2024-06-30" "2024-07-31" "2024-08-31" "2024-09-30"
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## [311] "2025-03-31" "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31"
## [316] "2025-08-31" "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
## [321] "2026-01-31" "2026-02-28" "2026-03-31"
```

#Reshape data

```
#Reshape Zillow city data from wide to long format
moco_long <- moco_cities %>%
  pivot_longer(
    cols = matches("^\\d{4}-\\d{2}-\\d{2}$|^X\\d{4}\\.\\d{2}\\.\\d{2}$"),
    names_to = "Date",
    values_to = "ZHVI"
  ) %>%
  mutate(
    Date = str_remove(Date, "^X"),
    Date = str_replace_all(Date, "\\.", "-"),
    Date = as.Date(Date)
  )
```

#Post cleaning inspection

```
glimpse(moco_long)
```

```
## Rows: 9,765
## Columns: 10
## $ RegionID   <int> 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081,…
## $ SizeRank   <int> 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79,…
## $ RegionName <chr> "Silver Spring", "Silver Spring", "Silver Spring", "Silver …
## $ RegionType <chr> "city", "city", "city", "city", "city", "city", "city", "ci…
## $ StateName  <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",…
## $ State      <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",…
## $ Metro      <chr> "Washington-Arlington-Alexandria, DC-VA-MD-WV", "Washington…
## $ CountyName <chr> "Montgomery County", "Montgomery County", "Montgomery Count…
## $ Date       <date> 2000-01-31, 2000-02-29, 2000-03-31, 2000-04-30, 2000-05-31…
## $ ZHVI       <dbl> 185818.2, 185642.1, 185688.0, 185978.9, 186870.6, 188282.9,…
```

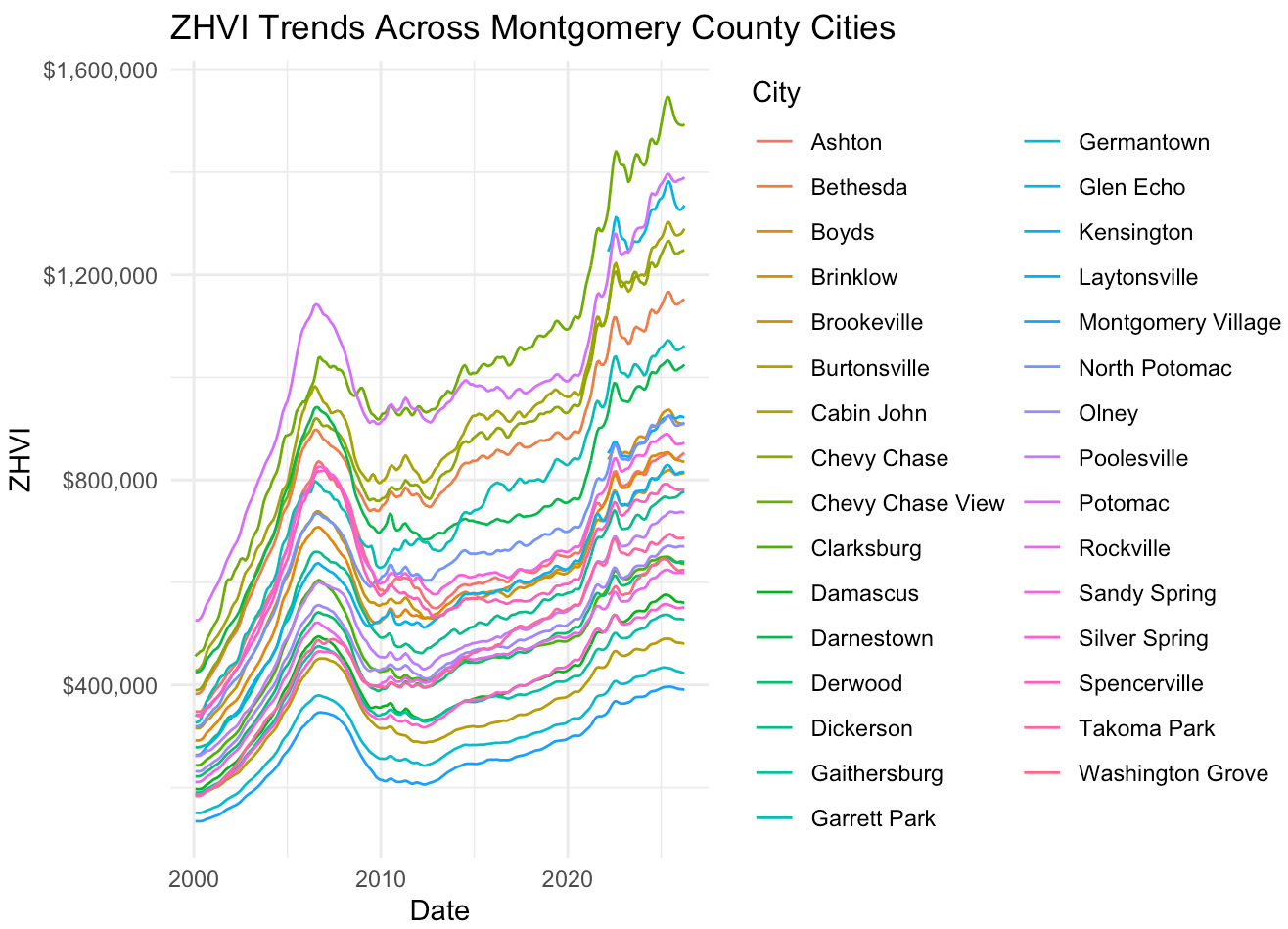
```
summary(moco_long$ZHVI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  134181  414426  564403  599572  756806 1547109    1060
```

#Time series visualization

```
#Home value trends across Montgomery County cities
ggplot(moco_long, aes(x = Date, y = ZHVI, color = RegionName)) +
  geom_line() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "ZHVI Trends Across Montgomery County Cities",
    x = "Date",
    y = "ZHVI",
    color = "City"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1060 rows containing missing values or values outside the scale rang
e
## (`geom_line()`).
```



#Latest prices calculation

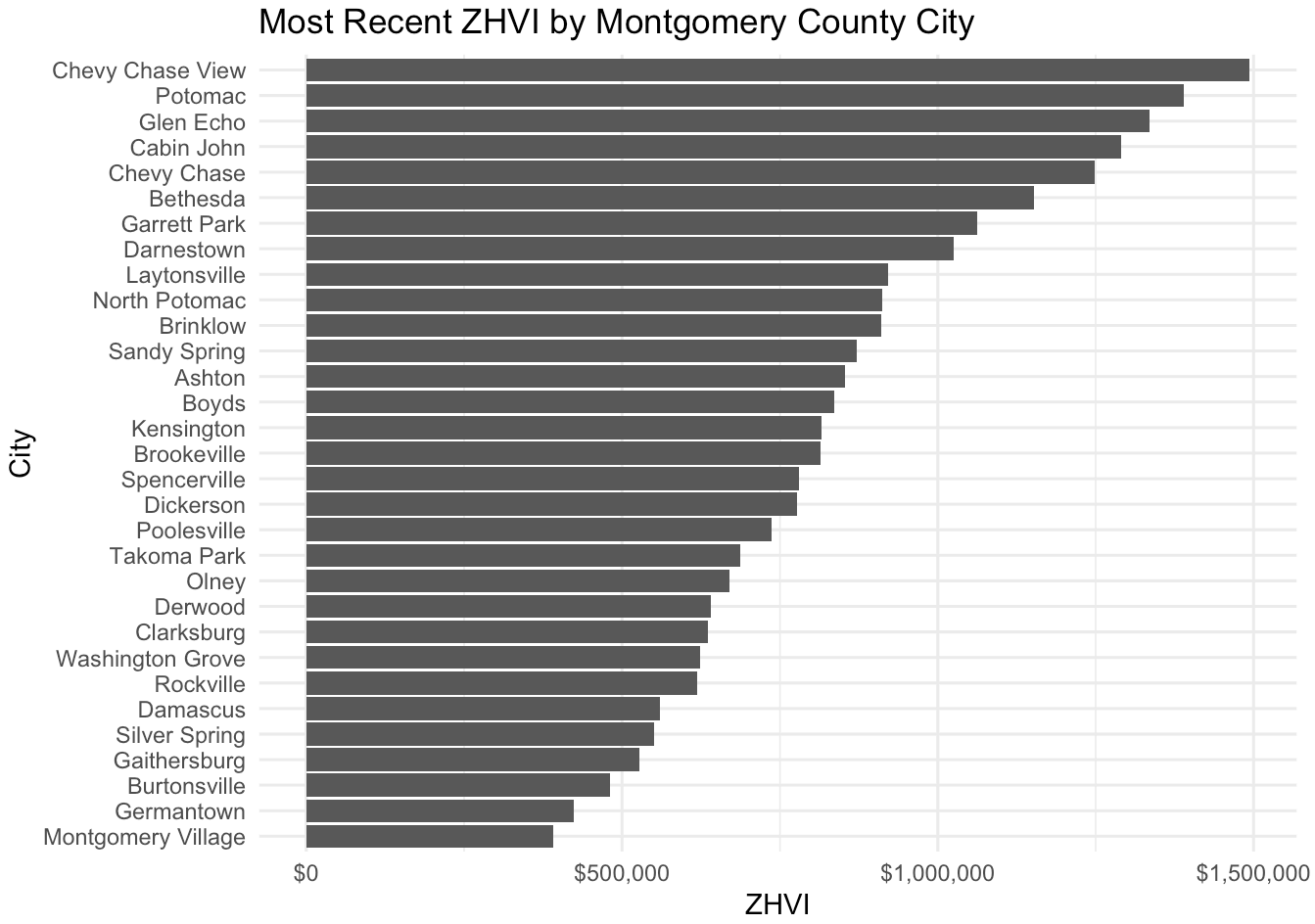
```
#Creating new Variable of latest pricing
latest_prices <- moco_long %>%
  group_by(RegionName) %>%
  filter(Date == max(Date, na.rm = TRUE)) %>%
  ungroup() %>%
  select(RegionName, Date, ZHVI) %>%
  arrange(desc(ZHVI))

latest_prices
```

```
## # A tibble: 31 × 3
##    RegionName       Date           ZHVI
##    <chr>            <date>        <dbl>
##  1 Chevy Chase View 2026-03-31 1493700.
##  2 Potomac          2026-03-31 1390212.
##  3 Glen Echo        2026-03-31 1335696.
##  4 Cabin John       2026-03-31 1289768.
##  5 Chevy Chase      2026-03-31 1247761.
##  6 Bethesda         2026-03-31 1152219.
##  7 Garrett Park     2026-03-31 1061769.
##  8 Darnestown       2026-03-31 1024744.
##  9 Laytonsville     2026-03-31  922014.
## 10 North Potomac    2026-03-31  911608.
## # ℹ 21 more rows
```

#Bar chart of latest prices

```
ggplot(latest_prices, aes(x = reorder(RegionName, ZHVI), y = ZHVI)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "Most Recent ZHVI by Montgomery County City",
    x = "City",
    y = "ZHVI"
  ) +
  theme_minimal()
```


Most Recent ZHVI by Montgomery County City
Chevy Chase View
Potomac
Glen Echo
Cabin John
Chevy Chase
Bethesda
Garrett Park
Darnestown
Laytonsville
North Potomac
Brinklow
Sandy Spring
Ashton
Boyds
Kensington
Brookeville
Spencerville
Dickerson
Poolesville
Takoma Park
Olney
Derwood
Clarksburg
Washington Grove
Rockville
Damascus
Silver Spring
Gaithersburg
Burtonsville
Germantown
Montgomery Village
City
$0
$500,000
$1,000,000
$1,500,000
ZHVI